

Family Pompilidae

Spider-hunting wasps

Occurrence 5,000 spp. worldwide, especially tropical and subtropical regions; wherever spider hosts are found



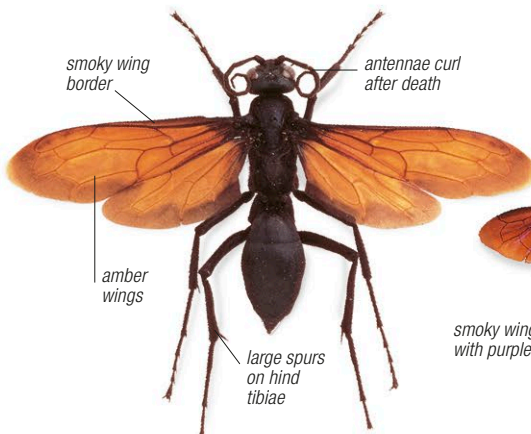
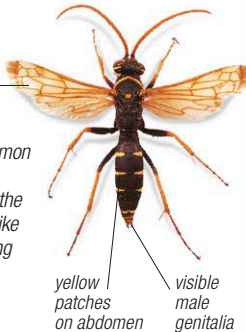
True to their name, these wasps can paralyze spiders with their strong venom. Females fly or run along the ground in search of spiders, drag the immobilized spider to a nest prepared in mud or in a crevice, and lay a single egg on it before

sealing the nest. Spider-hunting wasps have a painful sting. Most species are dark blue or black (although the wings may be dark yellow), and measure $\frac{3}{16}$ – $3\frac{1}{4}$ in (0.5–8 cm) long. Males are smaller than the females.

amber wings

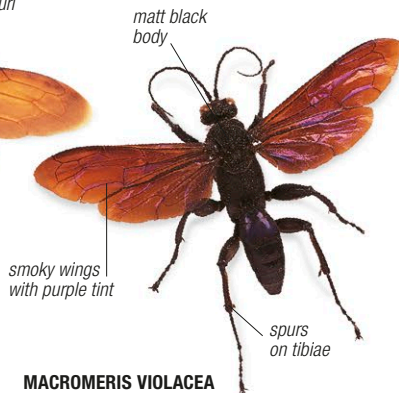
POMPIIUS SP.

This genus is common in warm regions, particularly where the ground is sandy. Like other spider-hunting wasps, *Pompilus* wasps are solitary.



PEPSIS HEROS

The tarantula hawk is the largest spider-hunting wasp, up to $3\frac{1}{4}$ in (8 cm) long. As its name suggests, it preys on tarantulas.



MACROMERIS VIOLACEA

The violet spider-hunting wasp, named for its iridescent purplish wings, is found in Southeast Asia. It is 2 – $2\frac{1}{4}$ in (5–5.5 cm) long.

Family Sphecidae

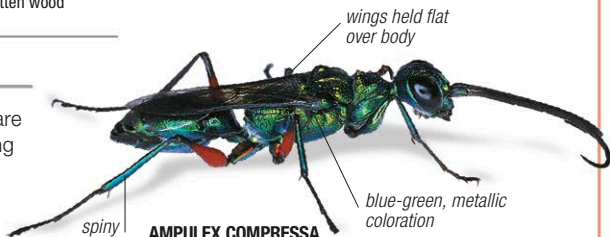
Digger wasps

Occurrence 8,000 spp. worldwide; in a variety of microhabitats—plant stems, soil, rotten wood



Some species of this family are also known as solitary hunting wasps, sand wasps, or mud dauber wasps. They are $\frac{5}{32}$ – $1\frac{1}{4}$ in (0.4–4.8 cm) long, relatively hairless, and often brightly colored. Digger wasps are solitary, and nest in plant stems,

soil, or rotten wood. The female catches an insect or a spider, paralyzes it, and buries it in a nest where she lays her egg. In this way, the emerging larva has a ready supply of food.



AMPULEX COMPRESSA

This African species, known as the emerald cockroach wasp, hunts cockroaches. It may enter houses in search of its quarry.

Family Scoliidae

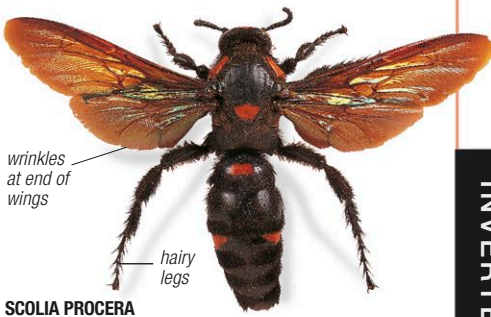
Mammoth wasps

Occurrence 500 spp. worldwide, especially in tropical regions; wherever scarab beetle hosts are found



As their name suggests, these are large wasps, measuring $\frac{3}{8}$ – $2\frac{1}{4}$ in (1–5.6 cm) in length, with stout bodies that are densely covered with hair. They are bluish black with reddish brown markings. Males are smaller and slimmer than females, with longer antennae. After mating, females hunt for the larvae of scarab beetles. They then sting, paralyze, and lay a single

egg on each larva. The wasp larva pupates in a cocoon that it spins inside the beetle larva's remains.



SCOLIA PROCERA

Found in Java, Borneo, and Sumatra, wherever scarab beetle larvae exist; this insect has a painful sting but is not very aggressive.

Family Vespidae

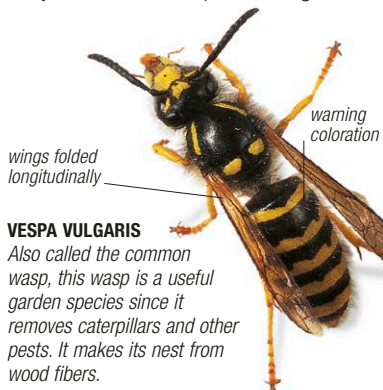
Social wasps

Occurrence 4,000 spp. worldwide; found in a variety of habitats, in nests and foraging widely for prey



The most familiar social wasps are the paper wasps and yellow jackets, which make nests out of chewed wood or other fibers. The queen overwinters, makes a nest in spring, and raises the first brood herself. Workers cooperate with her in caring for subsequent broods. Larvae develop in horizontal "combs," and are fed chewed-up

insects by sterile female workers. The adult wasps, which measure between $\frac{3}{8}$ and $1\frac{1}{2}$ in (0.4–3.6 cm) in length, roll or fold their wings longitudinally, rather than hold them flat over the body. Nearly all social wasps have warning coloration. They can also have a painful sting.



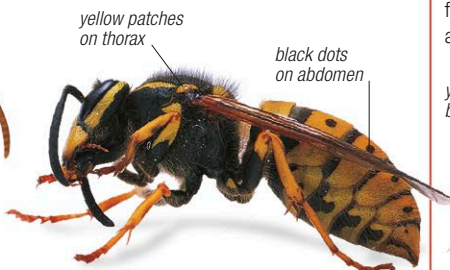
VESPA VULGARIS

Also called the common wasp, this wasp is a useful garden species since it removes caterpillars and other pests. It makes its nest from wood fibers.



VESPA CRABRO

Commonly known as the European hornet, this wasp makes its nest in hollow trees. Its colonies have only a few hundred workers.



VESPULE GERMANICA

Colonies of the German wasp may be perennial, with more than one queen. This wasp lives mostly in the Northern Hemisphere and was introduced to North America in the late 1800s.

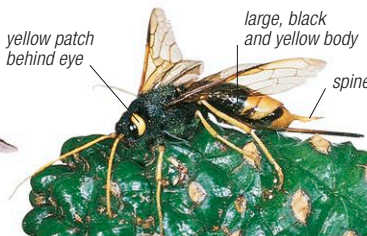
Family Siricidae

Horntails

Occurrence 150 spp. in temperate areas of the Northern Hemisphere; on and near deciduous and coniferous trees



Horntails or woodwasps are the largest members of the sawfly suborder (see right). Despite their menacing appearance, they do not sting—the "horn" at the end of their abdomen is a harmless spine. They are reddish brown, black and yellow, or metallic-purple, and measure $\frac{3}{4}$ – $1\frac{1}{2}$ in (1.8–4.2 cm) in length. Females have a long ovipositor and use it to lay their eggs inside tree trunks, in the process infecting them with fungi. Larvae burrow into heartwood and eat fungus and wood. Pupation occurs in a cocoon of silk and chewed wood.



UROCERUS GIGAS

The horntail shown above is a female common horntail or giant woodwasp. The male of this harmless species is elusive.

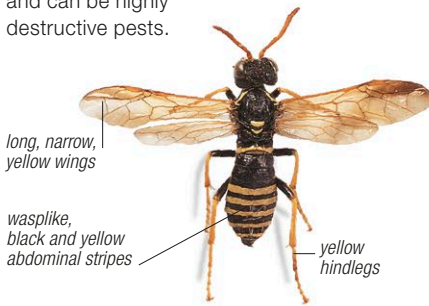
Family Tenthredinidae

Common sawflies

Occurrence 6,500 spp. worldwide, except New Zealand; on a wide range of plants in gardens, pastures, woodland



Sawflies often look like wasps, but are more primitive insects, classified in a suborder of their own. Unlike wasps, bees, and ants, they do not have a narrow waist, do not sting, and are solitary. Common sawflies are $\frac{1}{8}$ – $\frac{3}{4}$ in (0.3–2.2 cm) long. The females use their ovipositor to lay eggs in plants, and the caterpillar-like larvae, which are sometimes brightly colored, usually feed in the open on leaves. Common sawflies often attack fruit, vegetables, and trees, and can be highly destructive pests.



TENTHREDO SCROPHULARIAE

This figwort sawfly is a wasp-mimicking species, found in Europe and Asia. Its larvae feed on mullein and figwort plants.